

Brett Pauli, MSc

Passionate early-career aquatic ecologist with extensive field experience and over 8 years of data wrangling and statistical analysis of large ecological datasets.

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Education

Master of Science, Integrative Biology, University of Guelph, Advisors: Dr. Kevin McCann & Dr. Henri Vallès (2022- 2025)

- Integrated large datasets and implemented advanced statistical models and geospatial analyses.
- Thesis: *Environmental drivers of coral reef benthic states across scales*.
- Achieved 96% GPA and received Braithwaite Grant.
- Developed extensive knowledge in theoretical ecology, resource management and monitoring programs.
- Nominated for the Governor General Medal for distinguished research activities, teaching experience and leadership

Bachelor of Science (Honours), Major in Wildlife Biology and Conservation, Minor in Ecology, University of Guelph (2017- 2022)

- Dean's Honours List (five consecutive semesters).
- Relevant courses: Marine Ecological Processes (A), Population Genetics (A), Forest Biodiversity (A+), Wildlife Conservation and Management (A), Climate Change Biology (A-), Thesis in Integrative Biology (A).
- Thesis: *Shifting trait composition in boreal freshwater fish communities across gradients of anthropogenic stressors*.
- Completed two undergraduate independent research courses on (1) developing life-support systems for Mars missions and (2) assessing the effectiveness of natural resource policies in protecting threatened species.

Experience

Junior Ecologist – Sumac Environmental Consulting, Thornton, ON (current)

- Managed environmental projects from proposal preparation and budgeting to permitting, fieldwork, data analysis, GIS mapping, technical reporting, and client communication.
- Conducted pre-, during-, and post-construction environmental monitoring, including species at risk assessments, migratory bird nesting surveys, habitat assessments, fisheries and herpetofauna surveys, hydrometric monitoring, and compliance inspections (e.g., stormwater management and sediment and erosion controls).
- Interpreted environmental legislation and policy requirements to identify project constraints, develop mitigation measures, document non-compliance and corrective actions.
- Prepared GIS maps, photo documentation, technical reports, and environmental memoranda to support environmental compliance, while coordinating with construction crews, contractors, conservation authorities, municipal planners.

Research Contractor – Fisheries and Oceans Canada, Arctic Region, Arctic Fisheries and Marine Mammal Division (2026)

- Conducted statistical analyses and QA/QC, prepared stock assessment research report, generated publication-quality figures and digitized creel survey data.

Lab and Field Aquatic Research Assistant, Harkness Laboratory of Fisheries Research, Algonquin Park, ON (2022 –2023)

- Conducted extended periods of remote fieldwork in Algonquin Park ensuring safe completion of tasks while adapting to changing conditions and ensuring team safety and invasive species prevention.
- Sampled thousands of freshwater fish (e.g., *Cyprinid*, *Salmonidae*, etc.) and using numerous sampling techniques (e.g., electrofishing, acoustic telemetry, weirs, seine, fyke, gillnets, tooth nets, zooplankton hauls, etc.).
- Operated, trailered, launched research vessels (12-21ft) in challenging weather conditions.
- Collected plant and animal tissue for stable isotope analysis for key foodweb metrics.
- Extracted otoliths and collected water samples for chemistry analysis.
- Operated and maintained RBR loggers, YSI meters, GPS, trail camera, acoustic telemetry tagging equipment.
- Identified and handled dozens of freshwater fish and benthic macroinvertebrates (*Ephemeroptera*, *Trichoptera*).

Restoration Technician, Nottawasaga Valley Conservation Authority, Nottawasaga Watershed, ON (2021)

- Carried out river realignment and floodplain creation projects to restore key river habitat parameters.
- Conducted electrofishing fish rescues including relocation of dozens of at-risk Northern Brook Lamprey.
- Implemented standardized benthic invertebrate sampling (i.e. kick net surveys).
- Monitored water quality and conducted longitudinal hydro-morphological surveys before, during and after project.
- Planted thousands of trees, installed revetments, restored native prairie grassland. Conducted acoustic and visual bird surveys. Worked instream around heavy machinery.

Graduate Teaching Assistant, University of Guelph, ON (2023 – 2024)

- Led weekly seminars and tutorials for three different courses relating to Ecology, Aquatic Ecosystems, Biodiversity
- Guided students in both lab and field settings. Earned "Perfect" teaching team scores and student evaluations.
- Handled reptiles and amphibians, as well as assisted in lab dissections and implemented safety protocols.

Certifications & Additional Information

- Publications and conference presentations available upon request or at brettpauli99.github.io.
- Fluent in French & English (currently learning Spanish), Marathon Runner, Eligible for RPBio.
- Built machine learning models (YOLOv8) for identifying coral reef imagery (CVATai).
- PADI Open Water Diver with over a dozen dives spanning Honduras, Barbados, Hawaii, and Colombia.